Institute of Technology of Cambodia

Assignment 01

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Group A

I2-International Program

Lab-1: Basic Java syntax

Lab1-Ex1: Ask a user for a number, say n. Display all sequence numbers from 1000 to n. Find the sum and average of these numbers and display the result on screen.

```java
package assigment;
import java.util.Scanner;
public class ass_task1 {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        int num, sum=0;
        double average=0;
        System.out.print("Imput:");
        System.out.println("");

        Scanner sc = new Scanner(System.in);

        System.out.print("  Enter a number: ");
        num = sc.nextInt();

        System.out.println("");
        System.out.println("Output:");

        for (int i=1000 ; i>num ; i-- ) {
            System.out.print(i+" ");

            sum += i;
            average = sum / (1000-i+1);

        }

        System.out.println("");
        System.out.println("Sum is: "+sum);
        System.out.println("Average is: "+average);

    sc.close();


    }

}

```

```
<terminated> ass_task1 [Java Application] C:\Users\User\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_25.0.4.v20260826-
Imput:
       Enter a number: 168

Output:
1000 999 998 997 996 995 994 993 992 991 990 989 988 987 986 985 984 983 982 981 986
Sum is: 486304
Average is: 584.0
```

Lab1-Ex2: Solve a quadratic equation ax^2 + bx + c = 0

```java
package assigment;
import java.util.Scanner;

public class ass_task2 {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);

        int num1, num2, num3;
        double delta, x1, x2;

        System.out.println("=============================================");
        System.out.println("         Welcome to our program              ");
        System.out.println("'Quadratic Equation Solver ax^2 + bx + c = 0 ");
        System.out.println("=============================================");
        System.out.println("");
        System.out.println("");

        System.out.print("Input a: ");
        num1 = sc.nextInt();

        System.out.print("Input b: ");
        num2 = sc.nextInt();

        System.out.print("Input c: ");
        num3 = sc.nextInt();

        System.out.println("");
        System.out.println("");
        System.out.println("---------------------------------------------");

        System.out.print("Roots of this equation: "+num1+"x^2 +"+num2+"x +"+num3+" = 0");

        delta = (num2*num2) - 4*num1*num3;
        x1 = (-num2 + Math.sqrt(delta))/2*num1;
        x2 = (-num2 - Math.sqrt(delta))/2*num1;

        System.out.println();
        System.out.println("    X1= "+x1);
        System.out.println("    X2= "+x2);

        System.out.println("---------------------------------------------");
        sc.close();

    }


}
```

```
<terminated> ass_task2 [Java Application] C:\Users\User\.p2\pool\plugins\org.eclipse.justj.open
=============================================
         Welcome to our program
'Quadratic Equation Solver ax^2 + bx + c = 0
=============================================


Input a: 2
Input b: 5
Input c: 2


---------------------------------------------
Roots of this equation: 2x^2 +5x +2 = 0
        X1= -2.0
        X2= -8.0
---------------------------------------------
```

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Lab1-Ex3: Write a program that can ask users to input 5 numbers. The program processes the input from the user and find out the minimum and maximum numbers.

```java
package assigment;
import java.util.Scanner;

public class ass_task3 {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner sc = new Scanner(System.in);

        int num1,num2,num3,num4,num5, max=0,min=0;

        System.out.println("Input: ");
        num1 = sc.nextInt();
        num2 = sc.nextInt();
        num3 = sc.nextInt();
        num4 = sc.nextInt();
        num5 = sc.nextInt();

        max=num1;
        if(max<num2) {
            max=num2;

        }
        if(max<num3) {
            max=num3;
        }
        if(max<num4) {
            max=num4;
        }
        if(max<num5) {
            max=num5;
        }

        min=num1;
        if(min>num2) {
            min=num2;
        }
        if(min>num3) {
            min=num3;
        }
        if(min>num4) {
            min=num4;
        }
        if(min>num5) {
            min=num5;
        }



            System.out.println();
            System.out.println("Output:");
            System.out.println("Among all the five input numbers ("+num1+ ", "+num2+", "+num3+", "+num4+", "+num5+"),");
            System.out.print("min is "+min+" and max is "+max);

            sc.close();

        }


    }
```

```
<terminated> ass_task3 [Java Application] C:\Users\User\.p2\pool\plugins\org.eclipse.justj.openjdk.hot
Input:
99 35 4 6 47

Output:
Among all the five input numbers (99, 35, 4, 6, 47),
min is 4 and max is 99
```

Lab1-Ex4: Write a program that can check if the given input date (yyyy-mm-dd) is valid or invalid.

```java
package assigment;

import java.util.Scanner;

public class ass_task4 {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner input = new Scanner(System.in);
        System.out.print("Input: ");
        String date = input.nextLine();


        String[] parts = date.split("-");

        if (parts.length != 3) {
            System.out.println("Output: invalid");
            return;
        }

        int year = Integer.parseInt(parts[0]);
        int month = Integer.parseInt(parts[1]);
        int day = Integer.parseInt(parts[2]);

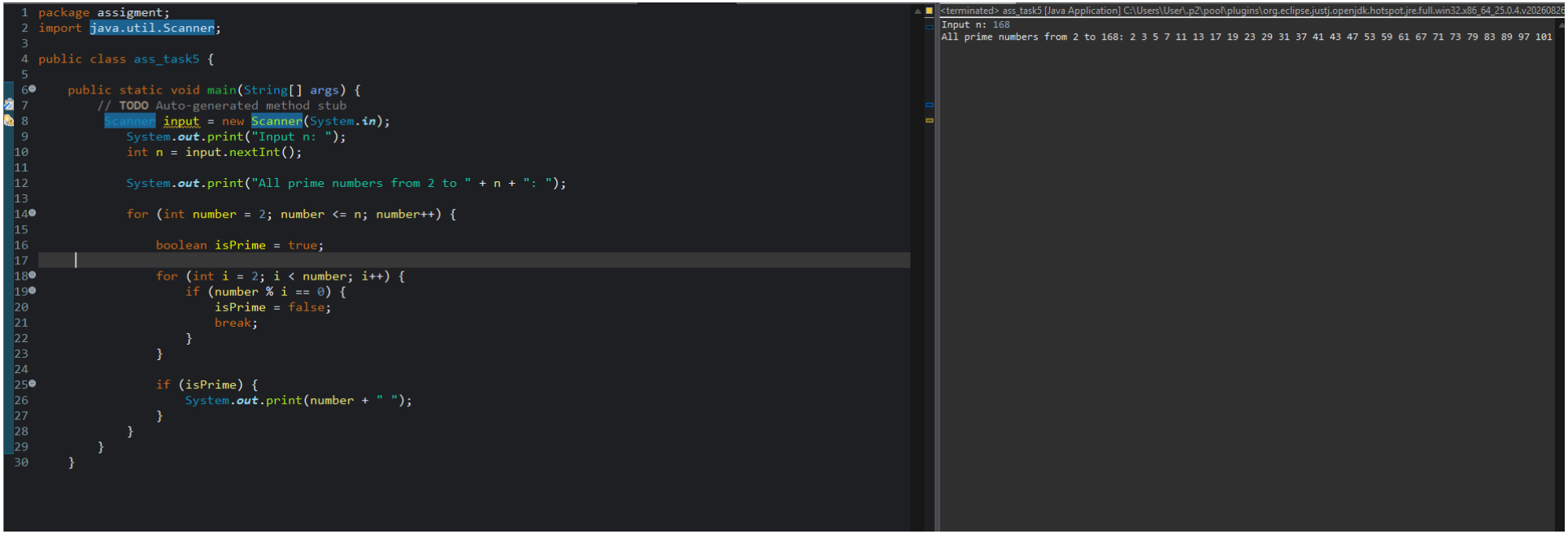
        boolean leapYear = (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);

        int daysInMonth;
        if (month == 2) {
            if (leapYear) {
                daysInMonth = 29;
            } else {
                daysInMonth = 28;
            }
        } else if (month == 4 || month == 6 || month == 9 || month == 11) {
            daysInMonth = 30;
        } else {
            daysInMonth = 31;
        }

        if (month < 1 || month > 12 || day < 1 || day > daysInMonth) {
            System.out.println("Output: invalid");
        } else {
            System.out.println("Output: valid");
        }
    }
}
```

```
<terminated> ass_task4 [Java Application] C:\Users\User\.p2\pool\plugins\org.eclipse.justj.openjdk.h
Input: 2026-12-30
Output: valid
```

Lab1-Ex5: Write a program that can display all prime numbers from 2 to n, where n is an input number by user.



```java
package assigment;
import java.util.Scanner;

public class ass_task5 {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner input = new Scanner(System.in);
           System.out.print("Input n: ");
           int n = input.nextInt();

           System.out.print("All prime numbers from 2 to " + n + ": ");

           for (int number = 2; number <= n; number++) {

               boolean isPrime = true;

               for (int i = 2; i < number; i++) {
                   if (number % i == 0) {
                       isPrime = false;
                       break;
                   }
               }

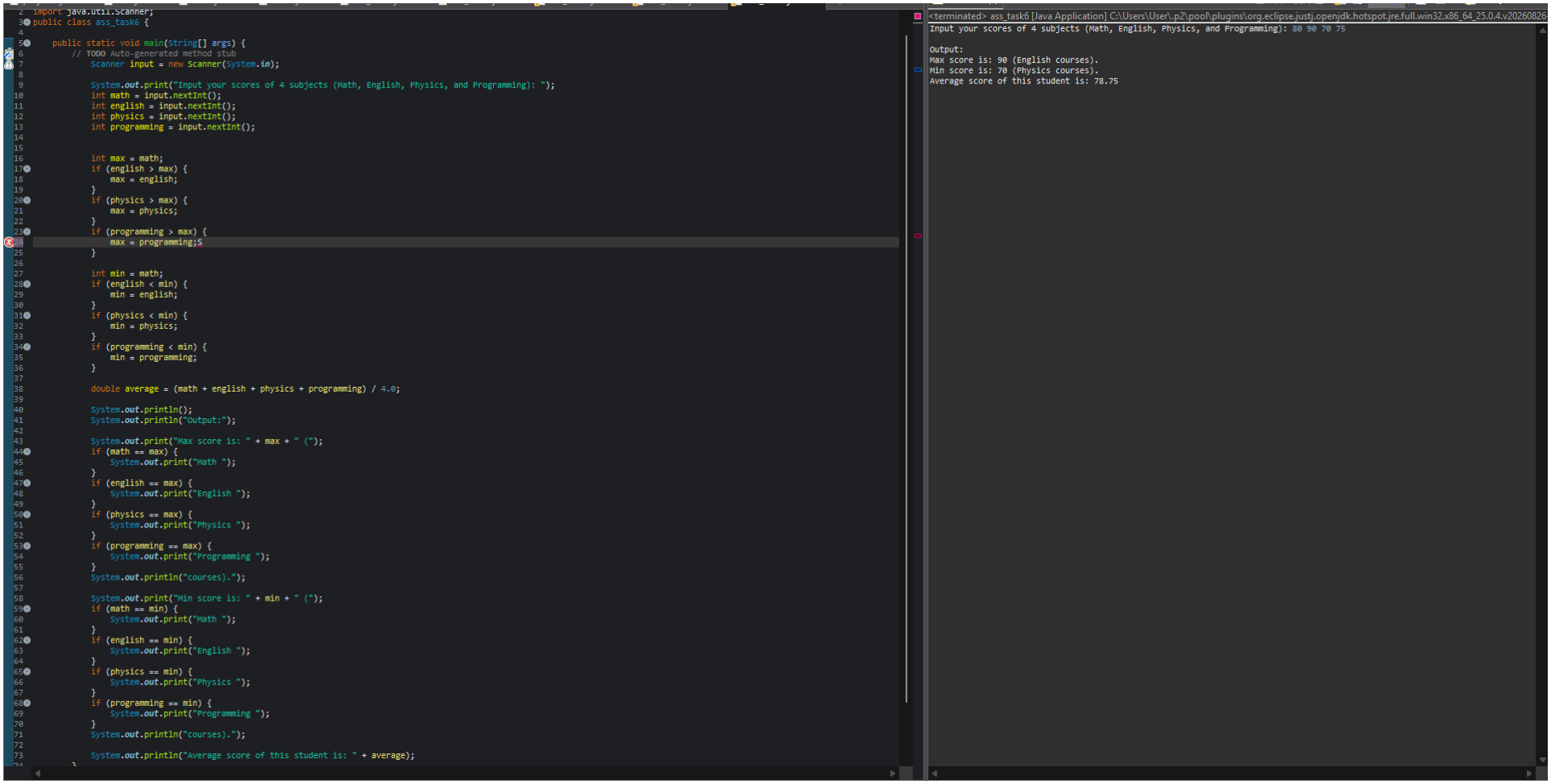
               if (isPrime) {
                   System.out.print(number + " ");
               }
           }
       }
    }
```

```
<terminated> ass_task5 [Java Application] C:\Users\User\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_25.0.4.v20260826
Input n: 168
All prime numbers from 2 to 168: 2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97 101
```

Lab1-Ex6: Write a program that prompt users for inputting scores of 4 subjects for Math, English, Physics, and Programming.

a. Which courses the students got max score, min score?

b. What are the average score of this student



```java
import java.util.Scanner;
public class ass_task6 {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Scanner input = new Scanner(System.in);

        System.out.print("Input your scores of 4 subjects (Math, English, Physics, and Programming): ");
        int math = input.nextInt();
        int english = input.nextInt();
        int physics = input.nextInt();
        int programming = input.nextInt();


        int max = math;
        if (english > max) {
            max = english;
        }
        if (physics > max) {
            max = physics;
        }
        if (programming > max) {
            max = programming;
        }

        int min = math;
        if (english < min) {
            min = english;
        }
        if (physics < min) {
            min = physics;
        }
        if (programming < min) {
            min = programming;
        }

        double average = (math + english + physics + programming) / 4.0;

        System.out.println();
        System.out.println("Output:");

        System.out.print("Max score is: " + max + " (");
        if (math == max) {
            System.out.print("Math ");
        }
        if (english == max) {
            System.out.print("English ");
        }
        if (physics == max) {
            System.out.print("Physics ");
        }
        if (programming == max) {
            System.out.print("Programming ");
        }
        System.out.println("courses).");

        System.out.print("Min score is: " + min + " (");
        if (math == min) {
            System.out.print("Math ");
        }
        if (english == min) {
            System.out.print("English ");
        }
        if (physics == min) {
            System.out.print("Physics ");
        }
        if (programming == min) {
            System.out.print("Programming ");
        }
        System.out.println("courses).");

        System.out.println("Average score of this student is: " + average);
    }
```

```
<terminated> ass_task6 [Java Application] C:\Users\User\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_25.0.4.v20260826
Input your scores of 4 subjects (Math, English, Physics, and Programming): 80 90 70 75

Output:
Max score is: 90 (English courses).
Min score is: 70 (Physics courses).
Average score of this student is: 78.75
```